

Lab 2: Agile Backlog Creation and Sprint Simulation in Jira

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Problem Statement: On-Demand Package Delivery Platform

Product Backlog: Epics and User Stories

Epic 1: Courier Booking and Cost Calculation

- **Jira Key:** SL-6
- **Requirement Source:** FR-001, FR-005
- **Description:** Senders book parcel pickups and the system calculates delivery prices based on parcel weight.

Story 1.1: Book Parcel Courier Pickup

- **Jira Key:** SL-9
- **Role:** As a sender client,
- **Action:** I want to enter pickup and destination addresses with package details,
- **Benefit:** So that I can book a delivery rider to transport my package.
- **Acceptance Criteria:** The system accepts pickup address, destination address, and weight, and then creates a booking record.
- **Priority:** High
- **Story Points:** 5

Story 1.2: Dynamic Cost Estimation by Weight

- **Jira Key:** SL-10
- **Role:** As a sender client,
- **Action:** I want to view the delivery price calculation based on package weight,
- **Benefit:** So that I know the accurate delivery cost before confirming the order.
- **Acceptance Criteria:** The system updates the total fare when package weight values change.
- **Priority:** High
- **Story Points:** 3

Epic 2: Intelligent Rider Matching and Dispatch

- **Jira Key:** SL-7
- **Requirement Source:** FR-003
- **Description:** The platform identifies active riders nearby and dispatches courier pickup tasks.

Story 2.1: Proximity-Based Rider Matching

- **Jira Key:** SL-11

- **Role:** As the dispatch platform,
- **Action:** I want to search for active delivery riders within a three-kilometer radius,
- **Benefit:** So that the system assigns the nearest available rider quickly.
- **Acceptance Criteria:** The system matches the order only with active riders within three kilometers of the pickup location.
- **Priority:** High
- **Story Points:** 5

Story 2.2: Rider Dispatch Task Notification

- **Jira Key:** SL-12
- **Role:** As a delivery rider,
- **Action:** I want to receive pickup requests with route instructions on my mobile app,
- **Benefit:** So that I can accept eligible delivery tasks.
- **Acceptance Criteria:** The system delivers a push notification with pickup address and package weight to the selected rider.
- **Priority:** High
- **Story Points:** 3

Epic 3: Live Telemetry Tracking and Destination OTP Verification

- **Jira Key:** SL-8
- **Requirement Source:** FR-002, FR-004
- **Description:** Senders track delivery riders in real time and delivery riders complete deliveries using recipient verification codes.

Story 3.1: Live GPS Telemetry Tracking

- **Jira Key:** SL-13
- **Role:** As a sender client,
- **Action:** I want to track the real-time GPS location of the assigned delivery rider on a live map,
- **Benefit:** So that I know the expected arrival time of my package.
- **Acceptance Criteria:** The system updates rider coordinates on the map with less than two seconds of latency.
- **Priority:** Medium
- **Story Points:** 5

Story 3.2: Destination OTP Verification

- **Jira Key:** SL-14
- **Role:** As a delivery rider,
- **Action:** I want to enter a six-digit one-time password provided by the recipient,
- **Benefit:** So that the system confirms package handover to the correct person.
- **Acceptance Criteria:** The system marks the delivery as completed only after successful code validation.
- **Priority:** Medium
- **Story Points:** 3

Product Backlog Summary Table

Jira Key	Epic Key	Epic Name	User Story Summary	Lab 1 Req	Priority	Story Points	Sprint
SL-9	SL-6	Courier Booking	Story 1.1: Book Parcel Courier Pickup	FR-001	High	5	Sprint 1
SL-10	SL-6	Courier Booking	Story 1.2: Dynamic Cost Estimation by Weight	FR-005	High	3	Sprint 1
SL-11	SL-7	Rider Dispatch	Story 2.1: Proximity-Based Rider Matching	FR-003	High	5	Sprint 1
SL-12	SL-7	Rider Dispatch	Story 2.2: Rider Dispatch Task Notification	FR-003	High	3	Sprint 1
SL-13	SL-8	Tracking & Verification	Story 3.1: Live GPS Telemetry Tracking	FR-002	Medium	5	Sprint 2
SL-14	SL-8	Tracking & Verification	Story 3.2: Destination OTP Verification	FR-004	Medium	3	Sprint 2

Sprint Execution and Simulation

Sprint 1: Booking Engine and Rider Dispatch

- **Sprint Goal:** Deliver parcel booking, weight cost calculation, and proximity rider dispatch.
- **Committed Story Points:** 16
- **Stories Completed:** SL-9 , SL-10 , SL-11 , SL-12
- **Summary:** The team moved all 16 story points through To Do, In Progress, and Done. The team completed Sprint 1 on schedule.

Sprint 2: Live Tracking and Secure Handover

- **Sprint Goal:** Deliver real-time GPS telemetry updates and destination OTP verification.
- **Committed Story Points:** 8
- **Stories Completed:** SL-13 , SL-14
- **Summary:** The team moved all 8 story points through To Do, In Progress, and Done. The team completed Sprint 2 on schedule.

Reflection Questions

1. Did your estimations reflect the actual effort?

I used the Fibonacci scale during planning poker. I assigned 5 story points to stories with algorithmic complexity, such as proximity rider matching and live map tracking. I assigned 3 story points to standard form validations and OTP verification logic. These estimates matched the work required during sprint execution.

2. Was your backlog well-prioritized?

Yes. I scheduled courier booking and rider dispatch in Sprint 1 because the platform needs working orders and assigned riders before anything else. I placed telemetry tracking and OTP verification in Sprint 2 to complete the post-dispatch delivery lifecycle.

3. How did your simulated sprint align with your plan?

I finished all planned user stories within the designated sprint timeframe. Sprint 1 completed 16 story points for the core booking engine. Sprint 2 completed 8 story points for live tracking and delivery verification without carrying incomplete items forward.

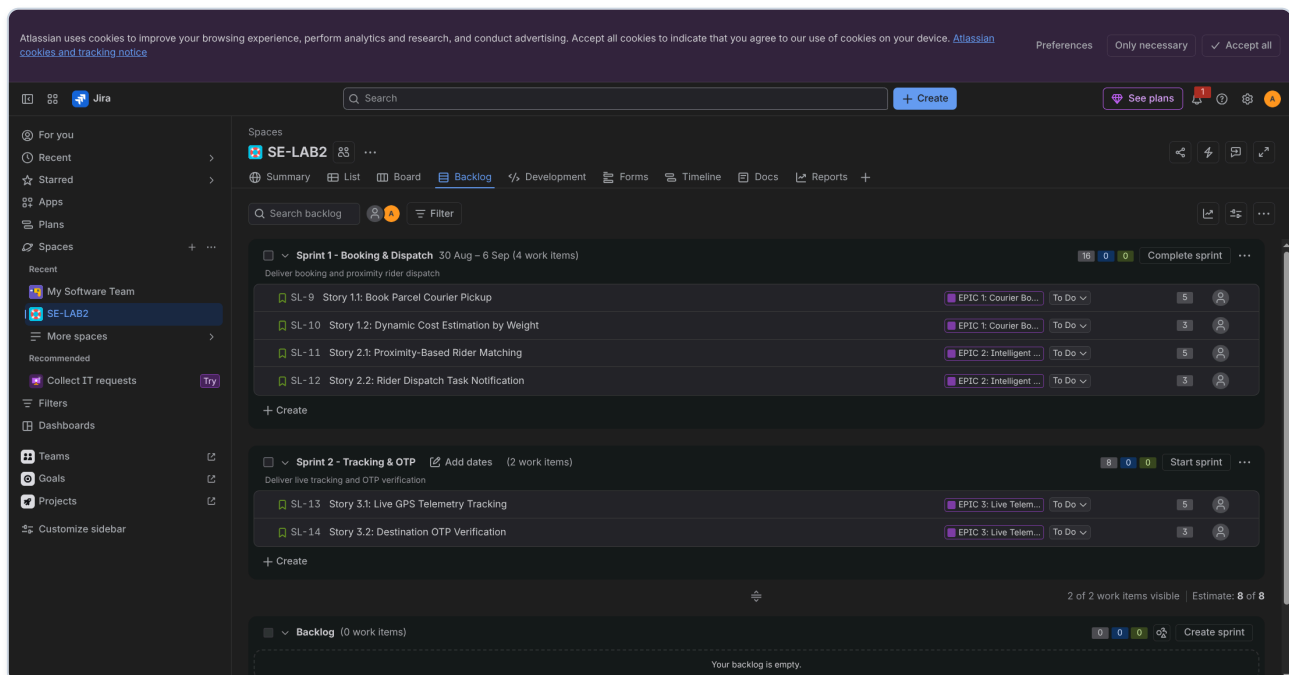
4. What insights did the burndown chart give about your team's capacity?

The burndown chart confirmed that I completed 16 story points in Sprint 1 and 8 story points in Sprint 2. This shows that I handle higher complexity during initial platform setup and maintains steady delivery speed across subsequent sprints.

Submission Screenshots

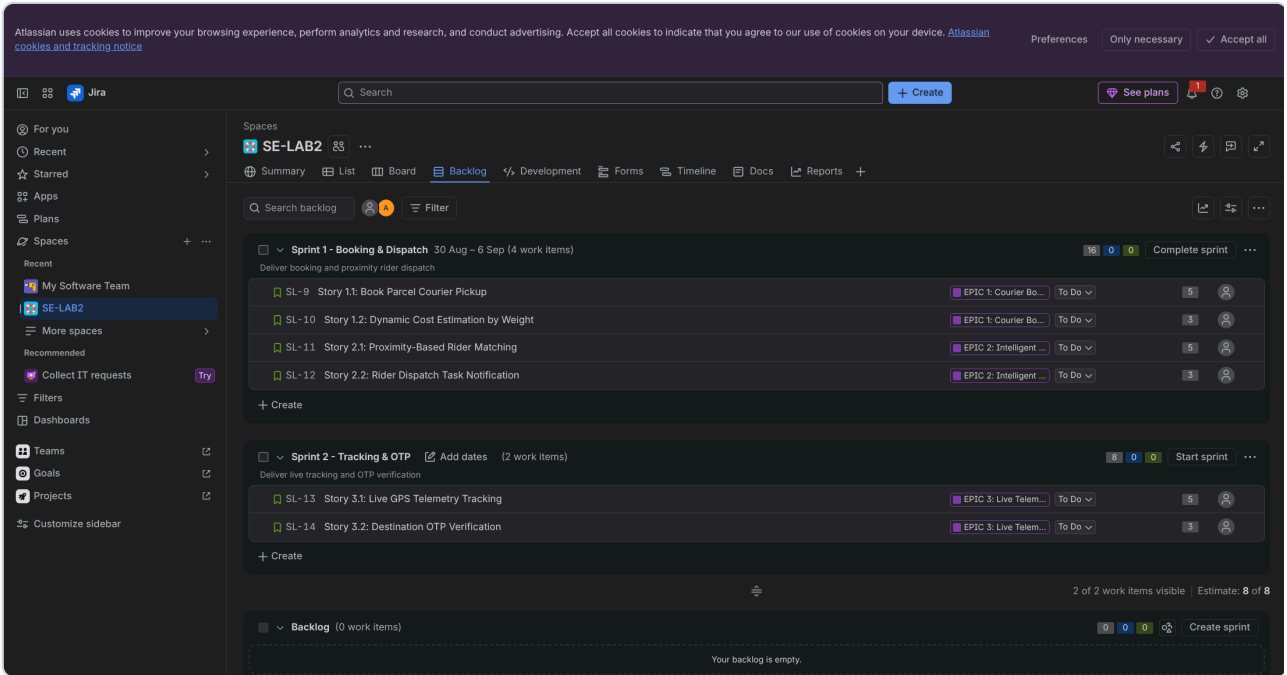
The following screenshots from Jira confirm the backlog setup, story points, active sprint board, and burndown metrics:

1. Jira Backlog and Epics View:



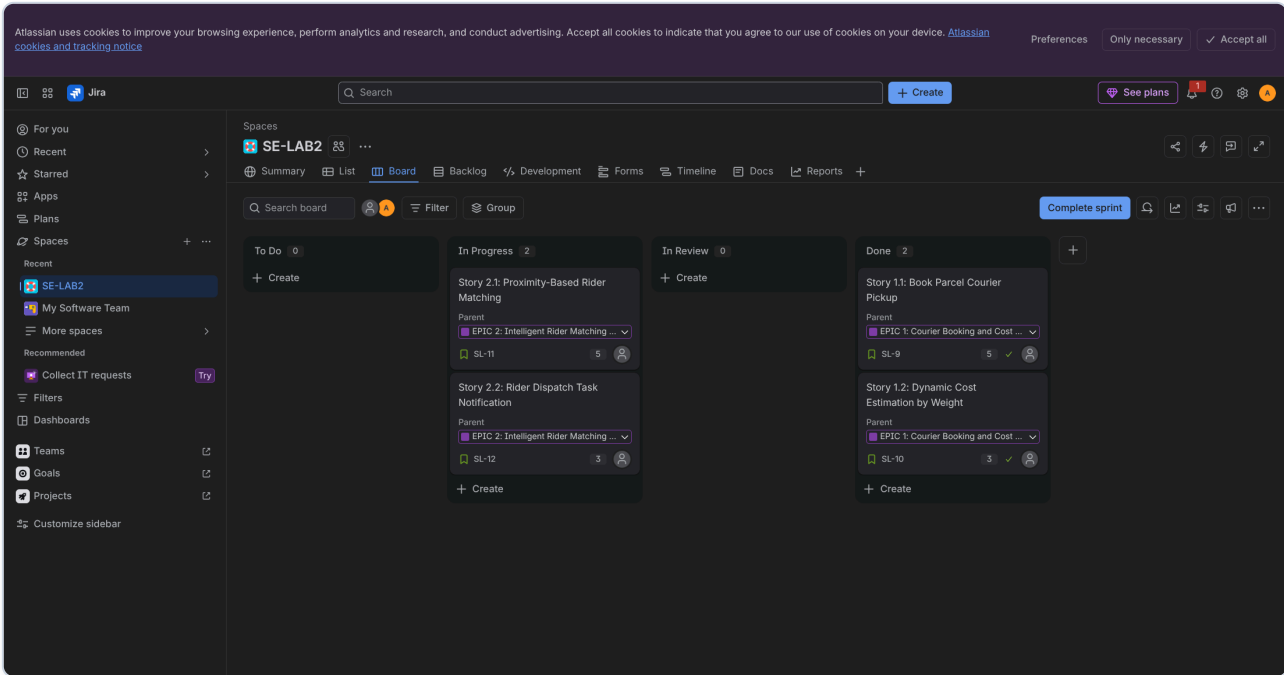
Shows the active backlog with Sprint 1 user stories, Epics, High priorities, and story points.

2. Full Backlog and Story Point Estimation:



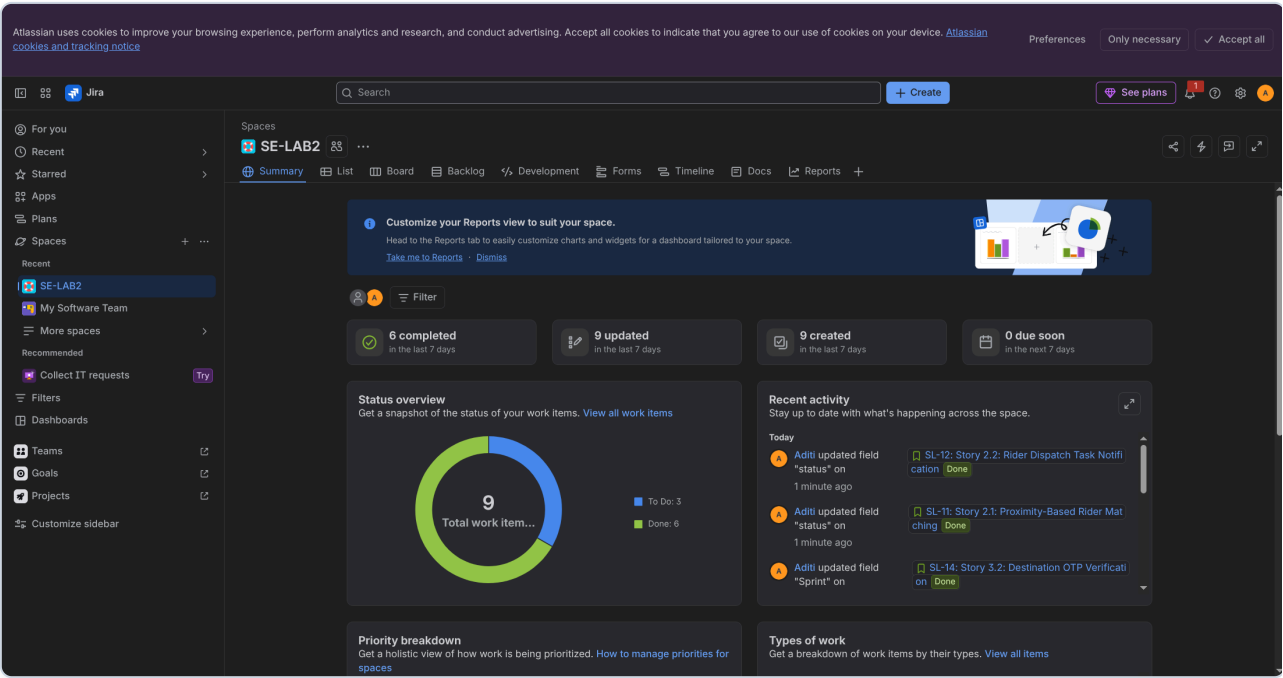
Shows the complete backlog with all 6 user stories across Sprint 1 (16 points) and Sprint 2 (8 points).

3. Active Sprint Board:



Shows the board during sprint execution with tasks transitioned across To Do, In Progress, and Done columns.

4. Burndown Chart and Project Reports:



Shows the completion metrics, work items by status, work items by type, and resolution trends.